



南 京 大 学 作 业 纸

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商数

题 8.2

$$1(2) \quad z = \sinh(x+y)$$

$$\frac{\partial z}{\partial x} = \sinh(x+y) + x \cosh(x+y)$$

$$\frac{\partial z}{\partial y} = x \cosh(x+y)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \cosh(x+y) + \cosh(x+y) - x \sinh(x+y) \\ = 2 \cosh(x+y) - x \sinh(x+y)$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = -x \sinh(x+y)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \cosh(x+y) - x \sinh(x+y)$$

$$1(4) \quad u = \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial u}{\partial x} = \frac{2}{3} \left(\frac{x}{y}\right)^{\frac{2}{3}-1} = \frac{2}{3} \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial u}{\partial y} = \frac{2}{3} \left(\frac{x}{y}\right)^{\frac{2}{3}-1} (-1) = -\frac{2}{3} \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial u}{\partial z} = \left(\frac{x}{y}\right)^{\frac{2}{3}} \ln \frac{x}{y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(2-1)}{x^2} \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{2(2+1)}{y^2} \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial^2 u}{\partial z^2} = \left(\frac{x}{y}\right)^{\frac{2}{3}} \ln^2 \frac{x}{y}$$

$$\frac{\partial^2 u}{\partial x \partial y} = -\frac{2}{xy} \left(\frac{x}{y}\right)^{\frac{2}{3}}$$

$$\frac{\partial^2 u}{\partial y \partial z} = \frac{1 + \ln \frac{x}{y}}{y}$$

$$\frac{\partial^2 u}{\partial z \partial x} = \frac{1 + \ln \frac{x}{y}}{x} \left(\frac{x}{y}\right)^{\frac{2}{3}} \quad \left(\frac{x}{y} > 0\right)$$

$$1(6) \quad z = \tan \frac{x}{y}$$

$$\frac{\partial z}{\partial x} = \frac{\frac{1}{y}}{\cos^2 \frac{x}{y}} = \frac{\sec^2 \frac{x}{y}}{y}$$

$$\frac{\partial z}{\partial y} = \frac{-x y^{-2}}{\cos^2 \frac{x}{y}} = -\frac{x}{y^2} \sec^2 \frac{x}{y}$$

$$\frac{\partial^2 z}{\partial x^2} = -2x y^{-3} \sec^2 \frac{x}{y} \tan \frac{x}{y}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -y^{-2} \sec^2 \frac{x}{y} - 2xy^{-3} \sec^2 \frac{x}{y} \tan \frac{x}{y}$$

$$\frac{\partial^2 z}{\partial y^2} = xy^{-3} \sec^2 \frac{x}{y} + 2xy^{-4} \sec^2 \frac{x}{y} \tan \frac{x}{y}$$

$$1(8) \quad z = e^{xy}$$

$$\frac{\partial z}{\partial x} = e^{xy} y \quad \frac{\partial z}{\partial y} = x e^{xy}$$

$$\frac{\partial^2 z}{\partial x^2} = y^2 e^{xy} \quad \frac{\partial^2 z}{\partial y^2} = x^2 e^{xy}$$

$$\frac{\partial^2 z}{\partial x \partial y} = (1+xy) e^{xy}$$

$$12. \text{证明: } f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0 = f(x,y) \big|_{(x,y)=(0,0)}$$

$\therefore f(x,y)$ 在 $(0,0)$ 连续.

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x} = 0$$

$$f_y(0,0) = 0$$

$$\text{取 } y = \Delta x, x = \Delta x, \text{ 则 } f(\Delta x, \Delta x) = \frac{\Delta x^2}{\sqrt{2}\Delta x} = \frac{\Delta x}{\sqrt{2}}$$

$$f(\Delta x, \Delta x) - f(0,0) = \frac{\Delta x \Delta x}{\sqrt{(\Delta x)^2 + (\Delta x)^2}} \quad \text{极限不存在.}$$

$\therefore$ 不可微

14. (1)

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{xy}}{x^2+y^2} \sin(x^2+y^2) = 0 = f(0,0)$$

$\therefore f(x,y)$ 在 $(0,0)$ 连续

$$\Delta z = f(\Delta x, \Delta y) - f(0,0)$$

$$= \frac{\sqrt{|\Delta x \Delta y|}}{(\Delta x^2 + \Delta y^2)} \sin[(\Delta x^2 + \Delta y^2)]$$

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = 0$$

$$\text{同理 } f_y(0,0) = 0$$

$$\Delta z = \Delta z - f_x(0,0)\Delta x - f_y(0,0)\Delta y$$

$$= \Delta z, \text{ 令 } \Delta y \rightarrow 0x$$

$$\text{则 } \lim_{\Delta x \rightarrow 0} \frac{\sqrt{(\Delta x)^2}}{\sqrt{2(\Delta x)^2}} \cdot \sin 2(\Delta x)^2 = \frac{1}{\sqrt{2}} \neq 0$$

$\therefore f(x,y)$ 在 $(0,0)$ 不可微

再证明偏导不连续

16. 证明:

$$f'_x(x,y) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 \cos \frac{1}{\Delta x^2}}{\Delta x}$$

$$= \Delta x \cos \frac{1}{\Delta x^2} = 0$$

$$f'_x(x,y) = 2x \cos \frac{1}{x^2 y^2} + x^2 y^2 \sin \frac{1}{x^2 y^2}$$

$$= 2x \left[\cos \frac{1}{x^2 y^2} + \frac{1}{x^2 y^2} \sin \frac{1}{x^2 y^2} \right]$$

极限为 0

但 y, kx 趋近于 0, 极限为 ∞ .

$\therefore$ 极限不同, 故不连续

$f'_x(x,y)$ 存在, 但 $f'_y(x,y)$ 不存在

$$\text{全微分 } b = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta z - f'_x(0,0)\Delta x - f'_y(0,0)\Delta y}{\sqrt{\Delta x^2 + \Delta y^2}}$$

$$= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta z}{\sqrt{\Delta x^2 + \Delta y^2}} = \frac{(\Delta x^2 + \Delta y^2) \cos \frac{1}{\Delta x^2 \Delta y^2}}{\sqrt{\Delta x^2 + \Delta y^2}}$$

$$= 0$$

$\therefore$ 可微.



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习题 8.3

$$1(1) u = f(x^2 + y^2 + z^2),$$

$$f'_x = 2xf'(x^2 + y^2 + z^2),$$

$$f''_{xx} = 2f'(x^2 + y^2 + z^2) + 4x^2 f''(x^2 + y^2 + z^2),$$

$$f''_{xy} = 4xy f''(x^2 + y^2 + z^2),$$

$$1(3) u = z^2 + y^2 + yz, z = \sin t, y = e^t$$

$$\frac{du}{dt} = \frac{\partial u}{\partial z} \frac{dz}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt}$$

$$= (2z + y) \cos t + (2y + z) e^t$$

$$= (2\sin t + e^t) \cos t + (2e^t + \sin t) e^t$$

$$1(5) z = x^2 y - xy^2, x = s \cos t, y = s \sin t$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$= (2xy - y^2) \cos t + (x^2 - 2xy) \sin t$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= s(y^2 - 2xy) \sin t + s(x^2 - 2xy) \cos t$$

$$7. \text{ 设 } u = f(x, y), v = g(x, y, u), w = h(x, u, v)$$

$$\frac{\partial w}{\partial x} = h'_1 + h'_2 \frac{\partial f}{\partial x} + h'_3 \left(g'_1 \frac{\partial f}{\partial x} \right)$$

$$= h'_1 + h'_2 \frac{\partial f}{\partial x} + h'_3 (g'_1 + g'_3 \frac{\partial f}{\partial x})$$

$$\frac{\partial w}{\partial y} = h'_2 \frac{\partial f}{\partial y} + h'_3 \left(g'_2 + g'_3 \frac{\partial f}{\partial y} \right)$$

$$= h'_2 \frac{\partial f}{\partial y} + h'_3 (g'_2 + g'_3 \frac{\partial f}{\partial y})$$

$$9. \text{ 设 } u = yf(\frac{x}{y}) + xg(\frac{y}{x})$$

$$\frac{\partial u}{\partial x} = y \cdot \frac{1}{y} f'(\frac{x}{y}) + g(\frac{y}{x}) + xy(-\frac{1}{x^2}) g'(\frac{y}{x})$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{y} f''(\frac{x}{y}) + \frac{1}{x^2} g'(\frac{y}{x}) + \frac{y}{x^2} g'(\frac{y}{x}) + \frac{y^2}{x^3} g''(\frac{y}{x})$$

$$= \frac{1}{y} f''(\frac{x}{y}) + \frac{y^2}{x^3} g''(\frac{y}{x})$$

$$\frac{\partial^2 u}{\partial x \partial y} = -\frac{x}{y^2} f'(\frac{x}{y}) - \frac{y}{x^2} g'(\frac{y}{x})$$

$$\Rightarrow x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = 0.$$

$$B. f(x, y) = \int_0^{xy} e^{-t^2} dt$$

$$\frac{x}{y} \frac{\partial^2 f}{\partial x^2} - 2 \frac{\partial^2 f}{\partial x \partial y} + \frac{y}{x} \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial f}{\partial x} = \frac{d}{dx} \int_0^{xy} e^{-t^2} dt = ye^{-x^2 y^2}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int_0^{xy} e^{-t^2} dt = xe^{-x^2 y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = -2xy^3 e^{-x^2 y^2}, \frac{\partial^2 f}{\partial y^2} = -2x^3 y e^{-x^2 y^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} (ye^{-x^2 y^2}) = e^{-x^2 y^2} - 2x^2 y^2 e^{-x^2 y^2}$$

$$\therefore \text{ 原式 } = \frac{x}{y} (-2xy^3 e^{-x^2 y^2}) - xe^{-x^2 y^2} - 2x^2 y^2 e^{-x^2 y^2}$$

$$+ \frac{y}{x} (-2x^3 y e^{-x^2 y^2}) = -2e^{-x^2 y^2}$$